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Magnetostratigraphy of the Niubao Formation in the Bangong-Nujiang Suture Zone: Constraints on the Amount of Crustal Shortening of the Tibetan Plateau Since 60 Ma

Shuai Li[1], Zhongpeng Han[2,3], Yalin Li[1,3], Xiaodong Tan[4,5], Alessandro Todrani[6], Xingduo Ma[1], Zijian Li[1], Xinhang Wang[1], Jie Dai[7], and Chengshan Wang[1,3]

[1]State Key Laboratory of Biogeology and Environmental Geology, School of Earth Sciences and Resources, China University of Geosciences (Beijing), Beijing, China, [2]Institute of Earth Sciences, China University of Geosciences (Beijing), Beijing, China, [3]Frontiers Science Center for Deep-time Digital Earth, China University of Geosciences, Beijing, China, [4]CAS Key Laboratory of Ocean and Marginal Sea Geology, South China Sea Institute of Oceanology, Chinese Academy of Sciences, Guangzhou, China, [5]Southern Marine Science and Engineering Guangdong Laboratory (Guangzhou), Guangzhou, China, [6]RINA Consulting S.p.A., Milan, Italy, [7]Key Laboratory of Sedimentary Basin and Oil and Gas Resources, Ministry of Natural Resources, Chengdu, China

Abstract The Early Cenozoic sedimentary archives of Tibet are crucial for elucidating the geodynamic processes related to continental collision, particularly crustal deformation and shortening. However, the lack of a reliable chronostratigraphic framework for the Cenozoic sedimentary basins in central Tibet has hindered a comprehensive understanding of the influence of the continuous India-Asia convergence process on the tectonic evolution of the Tibetan hinterland. Here, we report new magnetostratigraphic ages and paleomagnetic results for the Niubao Formation exposed in the northern bank of the Siling Co. Positive field tests and microscopic observations suggest that the remanence is primary. We determine that the depositional age of the measured section was between 59.2 and 48.6 Ma. Our results reveal that the paleogeographic location of the Bangong-Nujiang Suture Zone during the Late Paleocene–Early Eocene was 25.4 ± 2.1°N, thus supporting the model of two-stage collision between India and Asia.

Plain Language Summary The tectono-sedimentary evolution of the Cenozoic terrestrial sedimentary basins in the Tibetan Plateau was a long-term response to Cenozoic continental collision and was critical for both constraining the amount of crustal shortening of the Asian continent and for paleogeographic reconstruction of the plateau. Nonetheless, the ages of the basins are controversial because of the discordance between radiometric dating results and paleontological ages. Siling Co is the geographic boundary between the Lunpola and Nima Basins, and the Niubao Formation is widely exposed on its northeastern margin, providing excellent research materials for studying the paleogeographic evolution of the plateau. Hence, we established a reliable magnetostratigraphic sequence of the Niubao Formation in northwestern Siling Co, combined with U-Pb chronology, and constrained its age to 59.2–48.6 Ma. Our updated paleomagnetic results indicate that the paleolatitude of the Bangong-Nujiang Suture Zone was 25.4 ± 2.1°N during the Late Paleocene–Early Eocene. Combined with the published paleomagnetic data from the intra-oceanic arc, we suggest that the collision between the Asian and Indian continents had at least two stages and that the paleogeographic location of the collision was above 20°N.

Key Points:

- Detrital zircon U-Pb ages and paleomagnetic polarity patterns indicate that the Niubao Formation was deposited at 59.2–48.6 Ma
- The Bangong-Nujiang Suture Zone was located at 25.4 ± 2.1°N during the Paleocene–Eocene
- The amount of crustal shortening in the central Tibetan Plateau since 60 Ma is likely to be 780 ± 330 km

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

Z. Han and Y. Li,
hanzp@cugb.edu.cn;
liyalin@cugb.edu.cn

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Author Contributions:

Conceptualization: Xiaodong Tan, Chengshan Wang
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Methodology: Xiaodong Tan
Writing – original draft: Shuai Li, Zijian Li
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1. Introduction

The processes that have built the Tibetan Plateau have had far-reaching effects on continental-scale landscape shaping, global climate change, and biodiversity (Fang et al., 2020; Liu et al., 2019, 2023; Su et al., 2020; Xiong et al., 2022; Zhang et al., 2022). Previous investigations have demonstrated that central Tibet was the first region to uplift, and this uplift began in the Late Cretaceous (Wang et al., 2014). The region is generally regarded as consisting of the Lhasa and Qiangtang terranes, separated by the Bangong-Nujiang Suture Zone (BNSZ) (DeCelles et al., 2007; Kapp et al., 2005, 2007, Figure 1). Subsequently, the surrounding mountain ranges began to progressively uplift due to the collision between Asia and India during the Early Cenozoic, and this uplift accelerated the outward growth of the plateau and the concurrent inland shortening of Asia (Wang et al., 2008, 2014). Hence, defining the spatial and temporal processes of inland Asian shortening since the Early Cenozoic not

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only helps demonstrate the dynamics of continental collision but also provides constraints for reconstructing the paleogeographic evolution of the plateau (Kapp et al., 2005, 2007; Li et al., 2022; Tan et al., 2010).

Previous geological observations and paleomagnetic methodologies have estimated Cenozoic crustal shortening in Asia from >2,000 km to several hundred kilometers (Chen et al., 2010, 2014; Li et al., 2015, 2022; van Hinsbergen et al., 2011). Paleomagnetic methodologies are effective for quantitatively reconstructing the latitudinal movement of crustal blocks. However, distinct Cenozoic Asian paleogeographic positions, ranging from 30°N to the equator, have been determined due to the effects of inclination shallowing, remagnetization, and geomagnetic secular variation (Li et al., 2022). Consequently, these effects were responsible for the contentious amount of crustal shortening described above. Therefore, high-quality paleomagnetic data with reliable age constraints are urgently needed.

Kilometers-thick Paleocene to Miocene deposits were accumulated in several Cenozoic continental sedimentary basins along the BNSZ (DeCelles et al., 2007; Han et al., 2019; Kapp et al., 2005, 2007; Sun et al., 2014; Wei et al., 2017, Figure 1a). These sediments document how the paleogeographic evolution of uplifted central Tibet responded to the Cenozoic continental collision, and thus, the sediments are used as proxies of paleomagnetism to determine the amount of crustal shortening in central Tibet. However, although many recent investigations have established a well-dated Eocene to Miocene chronostratigraphic framework for these basins, that is, the Lunpola and Nima Basins (Fang et al., 2020; Han et al., 2019; Xiong et al., 2022), there is a lack of robust age constraints for the Early Cenozoic strata deposited close to the continental collisional period. This lack of age constraints hinders our ability to comprehensively understand how continental collision has impacted the crustal evolution of central Tibet.

The Nima and Lunpola Basins are the two largest basins within the BNSZ, and there is no distinct geological boundary between these basins, but Siling Co is the geographic boundary. On the northern shore of Siling Co, near Xiede village, the Paleocene–Eocene Niubao Formation is nearly 3 km thick. It is dominated by red fluvial and deltaic deposits, with a few lacustrine deposits in its upper part (Yin et al., 2021). Notably, the red beds in the Niubao Formation are substantially thicker in this lakeside area than in the Lunpola Basin, yet reliable age constraints are still lacking.

In this work, we report magnetostratigraphic and paleomagnetic data from the Tertiary strata in the northern lakeside region of Siling Co and combine them with detrital zircon U-Pb dating to determine the exact sedimentation age of the strata. Combined with the previous data, we constrain the paleogeographic positions of the Tibetan blocks and examine their responses to the Cenozoic continental collision and crustal shortening in Asia.

2. Geological Setting and Sampling

During or before the Early Cretaceous, the Qiangtang and Lhasa terranes collided and merged to produce the BNSZ, which spans approximately 2,000 km from west to east (Kapp et al., 2007; Wang et al., 2021). The studied red beds are located in a sedimentary depression between the Nima and Lunpola Basins. The depression stretches approximately 60 km in length and up to 15 km in width (Figure 1b). and is characterized by the folded Paleocene to Eocene Niubao Formation (Yin et al., 2021). We term this depression the Saibu Co depression, and it shares the tectonic framework with the two adjacent basins, which are bounded to the north by the north-dipping Shiquanhe-Gaize-Amdo Thrust and to the south by the south-dipping Gaize-Siling Co Thrust, which places strongly deformed Mesozoic strata in their hanging walls (Kapp et al., 2005).

The ca. 2,700 m thick measured section (hereinafter XDS) starts at 31°59′25.67″N, 88°37′26.79″E, and ends at 31°56′50.84″N, 88°37′44.96″E. Previous geological surveys conducted within the BNSZ identified plentiful ostracods and palynological assemblages in the Niubao Formation and yielded the Paleocene to Oligocene ages (Liao & Bai, 2019; Wang et al., 2009; Xia & Liu, 2008). This age range matches the stratigraphic age of the regional Niubao Formation that is constrained by insects, plants, and stratigraphic correlations (Liu et al., 2019; Yang et al., 2016; Zeng et al., 2020; Zhang et al., 2022; Zhao et al., 2020). Moreover, Su et al. (2020) constrained the age of the central part of the Niubao Formation to be 47.5 ± 0.7 Ma through fossil and radiometric dating, indicating that the central part of the Niubao Formation was at least of the Early Eocene in age, which is further supported by radiometric dating of the bentonite from the Niubao Formation (Zhao et al., 2020). However, the stratigraphic age of the Niubao Formation is still controversial (Fang et al., 2020; Xiong et al., 2022), especially in

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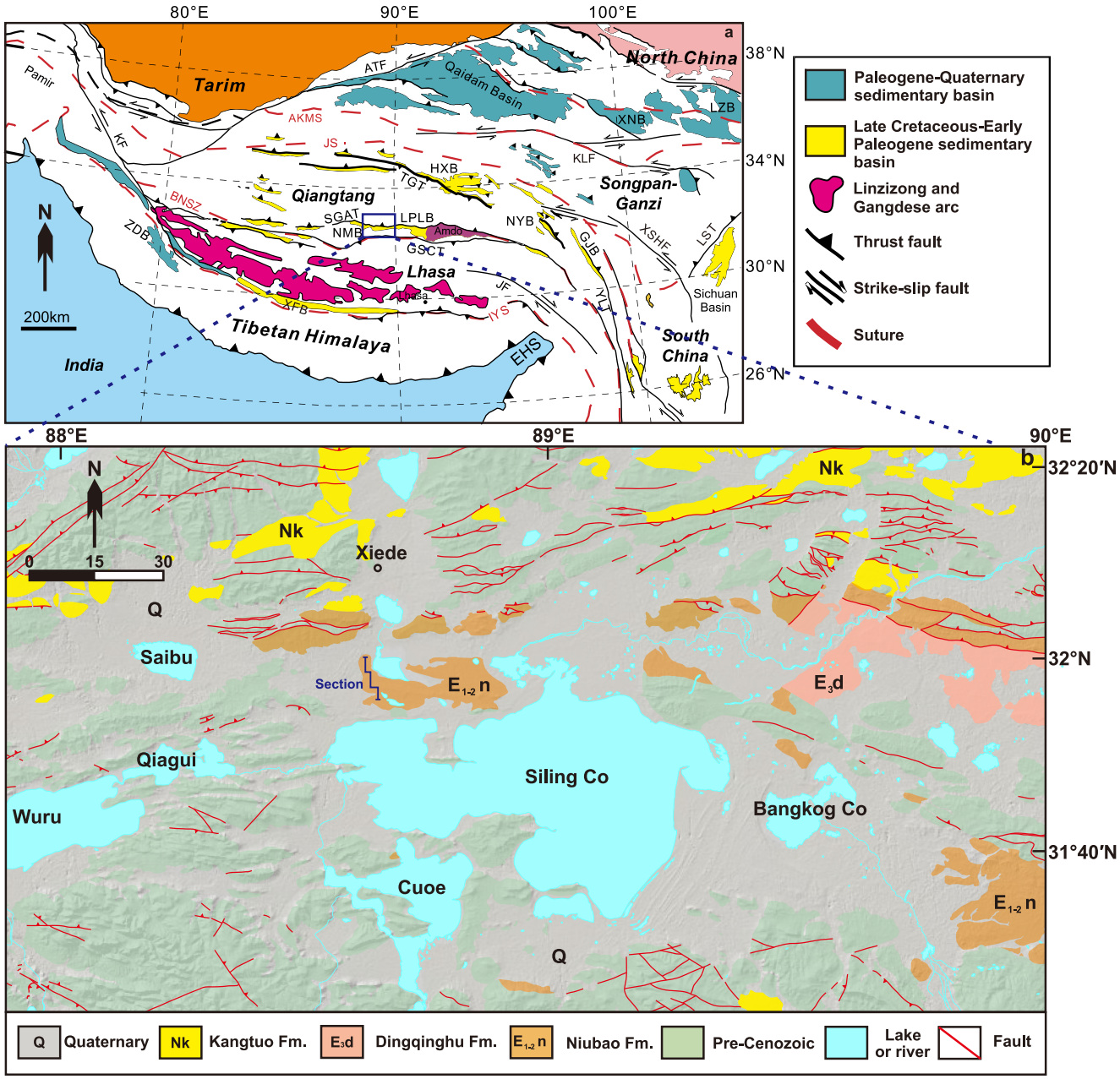


Figure 1. (a) Map showing the distribution of Late Cretaceous–Quaternary sedimentary basins in the Tibetan Plateau, modified from Li et al. (2020). The blue box represents our study area. The description of abbreviation can be seen in the supporting information. (b) Simplified geological map of the eastern edge of the Nima Basin, modified from Liu et al. (2019).

the Dayu section of the Lunpola Basin. Therefore, more reliable constraints of the Niubao Formation in new areas are needed to shed light on its depositional age.

With the help of a magnetic compass and a portable drill, we gathered and oriented 1,000 samples from the XDS section in accordance with the magnetostratigraphic sampling technique. The lithology of the samples was red sandstone and siltstone. Near the end of the XDS section is a syncline fold structure (Figure S2 in Supporting Information S1), where 100 samples were collected along the north limb of the fold for the paleomagnetic fold test. Two block samples of red sandstone were collected from the upper and middle parts of the section for zircon U-Pb dating (HD36U, HD44U; Figure S3 in Supporting Information S1).

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3. Results

3.1. Geochronology

In the upper and middle parts of the section, the youngest detrital zircon ages are 49.6 and 52.1 Ma, respectively. The numbers of reliable detrital zircons are 6 and 4 and thus can be used to constrain the maximum sedimentary age of the section (Dickinson & Gehrels, 2009).

3.2. Microscopic Characteristics of the Magnetic Minerals

Representative petrographic thin sections reveal that metallic minerals are black and opaque in plane and cross-polarized lights, while they appear light gray or off-white in reflected light. Scanning Electron Microscope (SEM) observations reveal that the metallic minerals range in size from 10 to 30 μm and are irregular in shape. They do not have oxidized edges, do not exist in cracks, and have a concave-convex contact relationship with peripheral non-metallic minerals (quartz, feldspar, etc.), which indicates that the magnetic carriers have not been remagnetized. Moreover, the elements that make up the metallic minerals are mainly iron and oxygen, and the semi-quantitative Fe/O atomic ratio is mostly 2/3, suggesting that the metallic minerals may be hematite (Figures S6–S8 in Supporting Information S1). The small amount of titanium may be an isomorph of hematite. Therefore, the magnetic carrier is detrital hematite, consistent with the unblocking temperature results and the decay curves of the detrital remanent magnetization intensity revealed by progressive thermal demagnetization in the next subsection.

3.3. Paleomagnetic Directions and Magnetostratigraphic Dating

We performed thermal demagnetization experiments on all collected samples. 769 out of 1,100 samples gave reliable paleomagnetic results, while the remaining were discarded from further considerations due to large uncertainties. Following a viscous component's isolation (define the interval of 30–200°C), a Characteristic Remanent Magnetization (ChRM) component was isolated in 696 samples, which were used for magnetostratigraphic considerations and the remaining 73 samples for the fold test (Figure 2). The remanent magnetic mineral is mainly composed of detrital hematite, according to microscopic observations and progressive thermal demagnetization results (Figures S4–S9 in Supporting Information S1). The direction of the Low Temperature Component (LTC) in *situ* coordinates is close to the current geomagnetic field (Declination = 0.5°, Inclination = 50.3°, Figure S10 in Supporting Information S1). The mean ChRM directions of the normal ($D = 6.9°$, $I = 35.1°$, $n = 374$) and the reversed ($D = 191.1°$, $I = -32.6°$, $n = 322$) polarities from the 696 samples are nearly antipodal (Figure S11 in Supporting Information S1). The angle ($\gamma = 4.3$) between the two mean directions is smaller than the threshold ($\gamma_c = 4.4$), indicating that the test is affirmative (Classification A) (McFadden & McElhinny, 1990). We converted the ChRM directions into Virtual Geomagnetic Pole latitudes and combined the directional data to determine the magnetostratigraphic sequence based on the measured thickness of the sampled horizons. Eventually, a total of 17 magnetic polarity zones were identified, including 9 reversed polarity zones (R1–R9) and 8 normal polarity zones (N1–N8) (Figure 3). Combined with the youngest Zircon U-Pb dating results, the magnetic polarity sequence correlates well with the global Geomagnetic Polarity Time Scale (GPTS) particularly in the Late Paleocene to Early Eocene, which was dominated by reversed polarity (Gradstein et al., 2012). The specific associated polarity zones are: N1 to C22n, R2 to C22r, N2–N3 to C23n, R4 to C23r, N4–N6 to C24n, R7 to C24r, N7 to C25n, R8 to C25r, N8 to C26n. Furthermore, we evaluated the relevance of the above magnetostratigraphic sequences using a systematic magnetostratigraphic correlation software based on the Dynamic Time Warping algorithm (Lallier et al., 2013). The results showed that the automatic polarity correction was consistent with the above polarity zones (Text S1, Figure S12 in Supporting Information S1). We thus conclude that the depositional age of the XDS section is between 59.2 and 48.6 Ma (Late Paleocene–Early Eocene).

To eliminate the influence of scattered paleomagnetic data, we used the Vandamme Cutoff method (Vandamme, 1994) to screen all ChRM directions, and finally obtained 730 filtered results. $D_g = 6.9°$, $I_g = -0.7°$, $k_g = 6.4$, $\alpha_{95} = 2.2°$ (geographic coordinate) and $D_s = 4.9°$, $I_s = 32.4°$, $k_s = 7.3$, $\alpha_{95} = 2.1°$ (stratigraphic coordinate) are the mean ChRM directions (Figure 2). The k_s/k_g of the ChRM mean directions is 1.13, larger than the critical value (1.11) (McElhinny, 1964). When the thresholds (31.43 and 44.44) and test results ($\xi_{I,g} = 75.60$ and $\xi_{I,s} = 31.89$) are compared, the fold test is likewise positive (McFadden, 1990). The Bootstrapped fold test is also positive (Tauxe & Watson, 1994; Figure S13 in Supporting Information S1). This suggests that the remanence

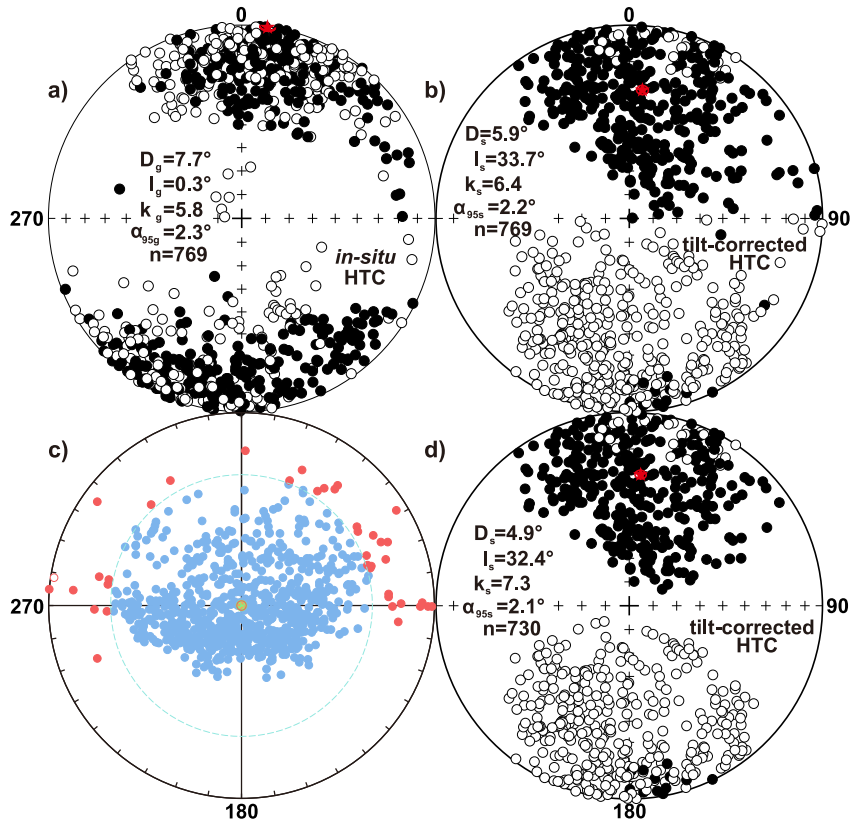


Figure 2. (a) and (b) Equal area projection of all the characteristic remanent magnetization (ChRM) directions in geographic and stratigraphic coordinate systems. The red five-pointed stars indicate the mean direction of the ChRM. (c) Distribution projection of virtual geomagnetic poles (VGPs) after screening by Vandamme Cutoff (Vandamme, 1994). Red dots represent VGPs that do not meet the constraints. (d) Equal area projection of the ChRM directions after removing the uncertain data.

were obtained before the fold. Furthermore, the 730 ChRM directions from the Niubao Formation were subjected to Elongation/Inclination (E/I) correction (Tauxe & Kent, 2004), and the optimal inclination was obtained between 41.0° and 46.0°, yielding a mean inclination of 43.5° and a correction factor of 0.58 (Figure S14 in Supporting Information S1).

4. Discussion and Conclusion

Our reliable paleomagnetic data and zircon U-Pb dating studies not only propose a depositional age of 59.2–48.6 Ma, providing a comprehensive and accurate age limit for the Early Cenozoic strata in central Tibet, which is different from the sedimentary sequence and age of the Lunpola Basin (Fang et al., 2020; Xiong et al., 2022) and is comparable to the regional Niubao Formation age (Su et al., 2020; Zhao et al., 2020, Text S1 in Supporting Information S1), but also constrain the BNSZ to be located at 25.4 ± 2.1°N during this period. Because the Lhasa and Qiangtang terranes had already collided during or before the Early Cretaceous (Wang et al., 2021), our paleolatitude may represent the boundary between the Lhasa and Qiangtang terranes. This is compatible with the paleomagnetic results of the Qiangtang terrane since the Late Cretaceous within the allowed error range (Chen et al., 2017; Huang et al., 1992; Lippert et al., 2011; Ma et al., 2023; Tong et al., 2015; Song et al., 2022, Figure 4a). The Late Cretaceous–Paleocene shortening of the Lhasa terrane is only 250 km (Li et al., 2015; van Hinsbergen et al., 2011). Considering that the present-day latitudinal range of the Lhasa terrane is about 3°, it can be deduced inversely that the southern edge of the Asian continent is about 20°N. However, another view is that the southern edge of Asia is located at northern hemisphere low-latitude positions (10°N or lower) (e.g., Chen et al., 2010, 2014; Yi et al., 2021), and there is a 2,000 km Greater Asia model. We summarize the paleomagnetic data from the Qiangtang and Lhasa terranes since 80 Ma in Table S1 in Supporting Information S1 (Reference point: 32°N and 88.5°E).

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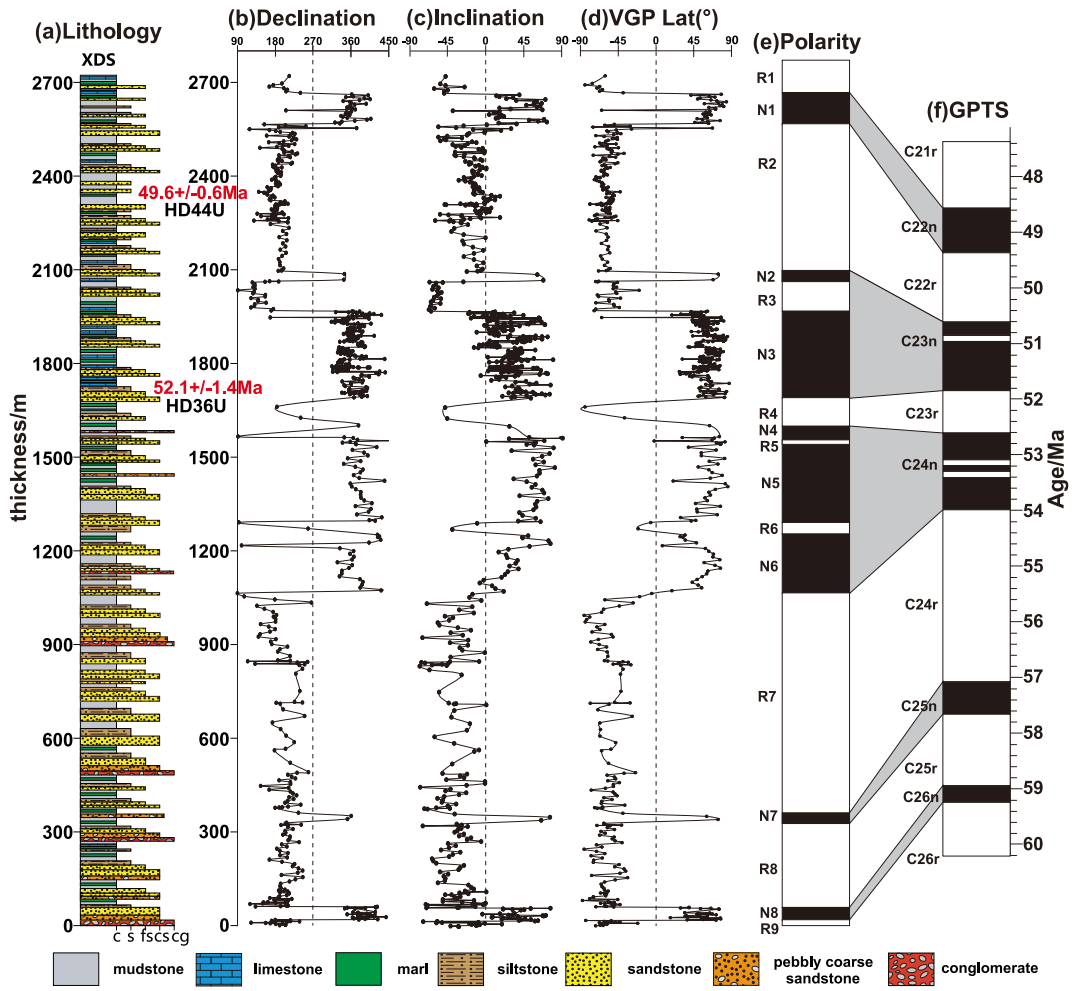


Figure 3. (a) Xiede South section lithological column (b)–(d) Variation diagram of declination inclination, and VGP with section thickness (e)–(f) Observed polarity of the section and the global Geomagnetic Polarity Time Scale (GPTS). Polarity correlations can be used to constrain stratigraphic ages (Gradstein et al., 2012).

The conflict of views is probably caused by the volcanic rocks without averaging out the geomagnetic secular variation, remagnetization and inclination shallowing during the paleomagnetic remanence acquisition processes (Li et al., 2022). First, Lippert et al. (2014) argued that paleomagnetic studies of individual and transiently erupted volcanic rocks may fail to eliminate the effects of the geomagnetic secular variation, and whether the obtained paleolatitudes can accurately define the paleogeographic position of the blocks is suspicious. Therefore, they conducted a more comprehensive evaluation of the paleomagnetic data of the Cenozoic volcanic rocks from the Lhasa terrane (Achache et al., 1984; Dupont-Nivet et al., 2010; Liebke et al., 2010; Sun et al., 2010; Tan et al., 2010), and selected a total of 41 sites. After averaging, the paleo-position of Asia should be around 20°N.

Second, Huang et al. (2015a, Huang et al., 1992, 2023) believed that thermal disturbance affected the primary remanence of the Dianzhong Formation (69–58 Ma) in the Linzhou area, while previous paleomagnetic directions (Chen et al., 2010, 2014; Yi et al., 2021) were overprinted by multi-period remagnetizations, which resulted in low inclinations and could not be used for paleogeographic reconstruction. In addition, the inclination shallowing of sedimentary rocks is also an important aspect that has to be considered in paleomagnetic studies. Huang et al. (2013) and Li et al. (2022) corrected the inclination at the top of the Linzizong Group and the Jialazi Formation in the Xigaze forearc basin, respectively. The results indicate that there is inclination shallowing, and the paleo-position calculated from the corrected inclination should be higher than 20°N.

Moreover, since the Cenozoic, the Lhasa terrane has not clearly experienced north-south tectonic deformation (Murphy et al., 1997; Yin & Harrison, 2000), so the inland shortening in Asia is mainly concentrated within and/

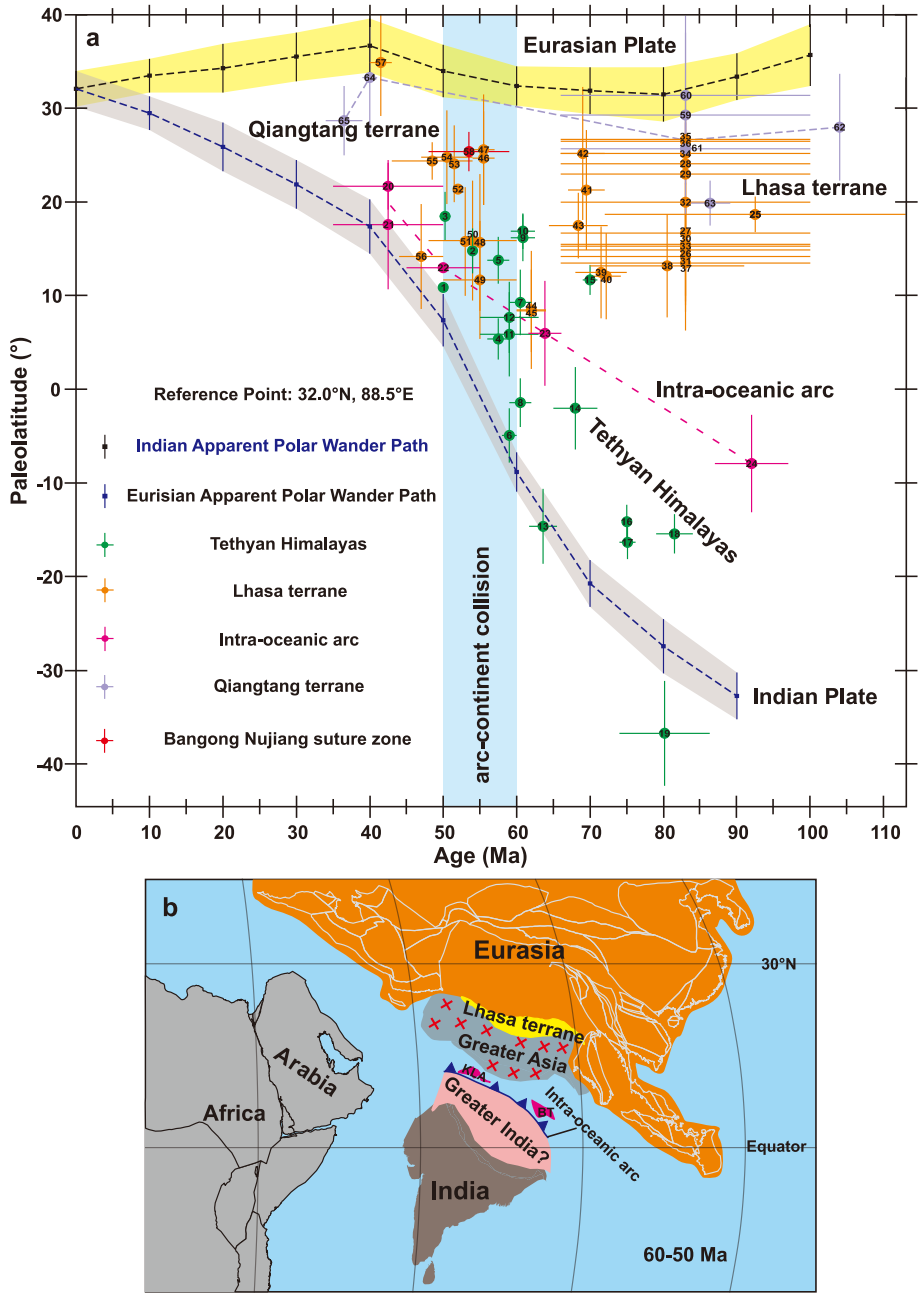


Figure 4. (a) Paleogeographic locations changes of the Eurasian plate, Indian plate, Lhasa and Qiangtang terranes since 70 Ma (modified from Li et al., 2022). The correspondence between the numbers and the paleomagnetic data is one-to-one (Table S1 in Supporting Information S1). The paleolatitudes of India and Eurasia are calculated according to Torsvik et al. (2012) with 32.0°N, 88.5°E as the reference point. (b) Paleogeographic reconstruction showing that India first collided with the intra-oceanic arc between 60 and 50 Ma, modified from Westerweel et al. (2019) and Martin et al. (2020). KLA, Kohistan-Ladakh arc; BT, Burma terrane.

or in the area north of the BNSZ. Furthermore, based on the variation of sediment accumulation rate (SAR) in the Niubao Formation, we recognized two different SAR stages, with an inflection point at 54 Ma, and the deposition rate increased from 204 to 297 m/Ma (Figure S15 in Supporting Information S1). This coincides with the timing of a sudden increase in SAR of the Hoh Xil Basin, which is likely a response to tectonic deformation in the central plateau and represents a far-field effect caused by plate convergence (Jin et al., 2018). In other words, the compressive stresses induced by the continuous north-south convergence between India and Eurasia were

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transmitted to the central plateau, which further led to the large-scale inland shortening of Asia during the Cenozoic (Jin et al., 2018; Yin & Harrison, 2000). Based on the 60 Ma Eurasian reference pole (Torsvik et al., 2012), the predicted paleolatitude is 32.5 ± 2.1°N, which has a paleolatitudinal difference of about 7.1 ± 3.0° from the observed paleolatitude (25.4 ± 2.1°N), suggesting that a north-south tectonic shortening of about 780 ± 330 km has occurred between the BNSZ and the Eurasian plate since the Late Paleocene–Early Eocene. This is consistent with the amount of inland shortening in Asia (~1,000 km) obtained from the comprehensive analysis of tectonic deformation, kinematic data, and other geological observations within the error range (Li et al., 2015; van Hinsbergen et al., 2011). Therefore, we suggest that the southern edge of Asia should be constrained to high-latitudes (20°N or higher) in the Early Cenozoic, and Greater Asia model should not exist.

At present, the collision model between India and Asia is mainly divided into a single-stage collision model (An et al., 2021; Hu et al., 2015) and a two-stage collision model (including the North India Sea model, Greater India Basin model and the arc-continent collision model) (Aitchison et al., 2000; van Hinsbergen et al., 2012, 2019; Yuan et al., 2020). Among them, the more popular view of the single collision model is that India and Asia collided at about 59 Ma (An et al., 2021; Hu et al., 2015). The new results of this study constrain the BNSZ to be at 25.4 ± 2.1°N during 59.2–48.6 Ma, and combined with the tectonic shortening of the Lhasa terrane, determine that the southern edge of Asia should be located above 20°N. Assuming that India and Asia collided at 59 Ma, the position of the Indian continent should be 8.8°S according to the apparent polar wander path of the Indian continent at 60 Ma (Torsvik et al., 2012).

Although the extent of Greater India is controversial (Besse et al., 1984; Bian et al., 2021; Dannemann et al., 2022; Jadoon et al., 2021a, 2024; Jin et al., 2023; Meng et al., 2020, 2012; Patzelt et al., 1996; Tong et al., 2008; Xu et al., 2022; Yang et al., 2019; Yi et al., 2011; Yuan et al., 2020, 2022, 2023; Zhang et al., 2017), it is patently absurd to compensate for a latitudinal extent of at least 28.8°, that is, the existence of a 3,200 km northern extension of the Greater India lithosphere. Furthermore, based on paleomagnetic research on the Burma terrane and Kohistan-Ladakh arc (Martin et al., 2020; Westerweel et al., 2019), recent studies suggest that there was an arc-continent collision at approximately 10°N around the Paleocene/Eocene boundary prior to the continent-continent collision, which is matched by a remarkable decrease in the convergence rate between Asian and Indian continents (Pusok & Stegman, 2020). At this time, the southern edge of Asia is above 20°N, which indicates that the Indian plate should collide with the intra-oceanic arc first and then the Asian continent during its northward drift. That is, the collision between India and Asia had at least two stages, but the extent and nature of Greater India (lithosphere or ocean basin) need more paleomagnetic data to confirm (Figure 4b). Moreover, identification of continental crust units such as the Longzi Block and the Zhongba Microterrane in southern Tibet indicates an archipelagic oceanic paleogeographic pattern in the Neo-Tethys Ocean, which also poses a challenge to the traditional India-Asia single collision model (Cheng et al., 2022; Xiao et al., 2022; Zhang et al., 2022). A multi-stage collision model is also supported by the modeling results of Liu et al. (2021) and seismic tomography (Van der Voo et al., 1999).

In conclusion, we suggest that Greater Asia model is not supported and the Early Tertiary paleogeographic position of Asia is likely situated at 20°N. Our results rule out the possibility that the continent-continent collision occurred at low latitudes close to the equator, and thus strongly support that the India-Asia collision was at least two stages involving an Early Tertiary arc-continent collision, and a later continent-continent collision.

Data Availability Statement

The raw paleomagnetic data, field outcrops, experimental results, and detailed descriptions of this article are available to everyone in Li (2024).

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References

Achache, J., Courtillot, V., & Zhou, Y. (1984). Paleogeographic and tectonic evolution of southern tibet since middle cretaceous time: New paleomagnetic data and synthesis. *Journal of Geophysical Research*, *89*(B12), 10311–10339. https://doi.org/10.1029/JB089iB12p10311

Aitchison, J. C., Badengzhu, Davis, A. M., Liu, J., Luo, H., Malpas, J. G., et al. (2000). Remnants of a Cretaceous intra-oceanic subduction system within the Yarlung–Zangbo suture (southern Tibet). *Earth and Planetary Science Letters*, *183*, 231–244. https://doi.org/10.1016/S0012-821X(00)00287-9

An, W., Hu, X., Garzanti, E., Wang, J., & Liu, Q. (2021). New precise dating of the India-Asia collision in the Tibetan Himalaya at 61 Ma. *Geophysical Research Letters*, *48*(3), e2020GL090641. https://doi.org/10.1029/2020gl090641

Besse, J., Courtillot, V., Pozzi, J. P., Westphal, M., & Zhou, Y. X. (1984). Paleomagneitc estimates of crustal shortening in the Himalayan thrusts and Zangbo suture. *Nature*, *311*(5987), 621–626. https://doi.org/10.1038/311621a0

Bian, W., Yang, T., Peng, W., Wang, S., Gao, F., Zhang, S., et al. (2021). Paleomagnetic constraints on the India-Asia collision and the size of greater India. *Journal of Geophysical Research: Solid Earth*, *126*(6), e2021JB021965. https://doi.org/10.1029/2021jb021965

Chen, J., Huang, B., & Sun, L. (2010). New constraints to the onset of the India-Asia collision: Paleomagnetic reconnaissance on the Linzizong Group in the Lhasa block, China. *Tectonophysics*, *489*(1–4), 189–209. https://doi.org/10.1016/j.tecto.2010.04.024

Chen, J., Huang, B., Yi, Z., Yang, L., & Chen, L. (2014). Paleomagnetic and $^{40}Ar/^{39}Ar$ geochronological results from the Linzizong Group, Linzhou basin, Lhasa terrane, tibet: Implications to paleogene paleolatitude and onset of the India-Asia collision. *Journal of Asian Earth Sciences*, *96*, 162–177. https://doi.org/10.1016/j.jseaes.2014.09.007

Chen, W., Zhang, S., Ding, J., Zhang, J., Zhao, X., Zhu, L., et al. (2017). Combined paleomagnetic and geochronological study on Cretaceous strata of the Qiangtang terrane, central Tibet. *Gondwana Research*, *41*, 373–389. https://doi.org/10.1016/j.gr.2015.07.004

Cheng, J., Li, Y., Li, S., Xiao, S., Bi, W., & Zou, Y. (2022). Reconstruction of the south Qiangtang–Zhongba–Tethyan Himalaya continental margin system along the northern Indian plate: Insights from the paleobiogeography of the Zhongba microterrane. *Journal of Asian Earth Sciences*, *240*, 105376. https://doi.org/10.1016/j.jseaes.2022.105376

Dannemann, S., Appel, E., Rösler, W., Neumann, U., Liebke, U., & Nag, D. (2022). Palaeomagnetic indication for India-Asia collision at 12°N and maximum 810 km Greater India extent in the western suture zone. *Geophysical Journal International*, *229*(2), 1193–1211. https://doi.org/10.1093/gji/ggab528

DeCelles, P. G., Kapp, P., Ding, L., & Gehrels, G. E. (2007). Late Cretaceous to middle Tertiary basin evolution in the central Tibetan Plateau: Changing environments in response to tectonic partitioning, aridification, and regional elevation gain. *Geological Society of America Bulletin*, *119*(5–6), 654–680. https://doi.org/10.1130/b26074.1

Dickinson, W. R., & Gehrels, G. E. (2009). Use of U-Pb ages of detrital zircons to infer maximum depositional ages of strata: A test against a Colorado plateau Mesozoic database. *Earth and Planetary Science Letters*, *288*(1–2), 115–125. https://doi.org/10.1016/j.epsl.2009.09.013

Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., Meijers, M. J. M., & Kapp, P. (2010). Palaeolatitude and age of the Indo-Asia collision: Palaeomagnetic constraints. *Geophysical Journal International*, *182*(3), 1189–1198. https://doi.org/10.1111/j.1365-246X.2010.04697.x

Fang, X., Dupont-Nivet, G., Wang, C., Song, C., Meng, Q., Zhang, W., et al. (2020). Revised chronology of central tibet uplift (Lunpola Basin). *Science Advances*, *6*(50), eaba7298. https://doi.org/10.1126/sciadv.aba7298

Gradstein, F., Ogg, J., Schmitz, M., & Ogg, G. (2012). *The geologic time scale 2012*. Elsevier B. V.

Han, Z., Sinclair, H. D., Li, Y., Wang, C., Tao, Z., Qian, X., et al. (2019). Internal drainage has sustained low-relief Tibetan landscapes since the early Miocene. *Geophysical Research Letters*, *46*(15), 8741–8752. https://doi.org/10.1029/2019GL083019

Hu, X., Garzanti, E., Moore, T., & Raffi, I. (2015). Direct stratigraphic dating of India-Asia collision onset at the Selandian (middle Paleocene, 59 ± 1 Ma). *Geology*, *43*(10), 859–862. https://doi.org/10.1130/g36872.1

Huang, K., Opdyke, N. D., Li, J., & Peng, X. (1992). Paleomagnetism of cretaceous rocks from eastern Qiangtang terrane of tibet. *Journal of Geophysical Research*, *97*(B2), 1789–1799. https://doi.org/10.1029/91jb02747

Huang, W., Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., Dekkers, M. J., Guo, Z., et al. (2015a). Can a primary remanence be retrieved from partially remagnetized Eocene volcanic rocks in the Nanmulin Basin (southern Tibet) to date the India-Asia collision? *Journal of Geophysical Research: Solid Earth*, *120*(1), 42–66. https://doi.org/10.1002/2014jb011599

Huang, W., Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., Dekkers, M. J., Waldrip, R., et al. (2015b). What was the paleogene latitude of the Lhasa terrane? A reassessment of the geochronology and paleomagnetism of Linzizong volcanic rocks (Linzhou basin, tibet). *Tectonics*, *34*(3), 594–622. https://doi.org/10.1002/2014tc003787

Huang, W., Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., & Hallot, E. (2013). Inclination shallowing in Eocene Linzizong sedimentary rocks from southern tibet: Correction, possible causes and implications for reconstructing the India-Asia collision. *Geophysical Journal International*, *194*(3), 1390–1411. https://doi.org/10.1093/gji/ggt188

Huang, W., Lippert, P. C., Reiners, P. W., Quade, J., Kapp, P., Ganerød, M., et al. (2022). Hydrothermal events in the Linzizong Group: Implications for paleogene exhumation and paleoaltimetry of the southern Tibetan plateau. *Earth and Planetary Science Letters*, *583*, 117390. https://doi.org/10.1016/j.epsl.2022.117390

Huang, W., Niu, S., Dekkers, M. J., Lippert, P. C., Bilardello, D., Solheid, P., et al. (2023). Remagnetization under hydrothermal alteration of south Tibetan Paleocene lavas: Maghemitization, hematization, and grain size reduction of (Titano)magnetite. *Journal of Geophysical Research: Solid Earth*, *128*(3), e2023JB026418. https://doi.org/10.1029/2023jb026418

Jadoon, U. F., Huang, B., Shah, S. A., Rahim, Y., Khan, A. A., & Bibi, A. (2021a). Multi-stage India-Asia collision: Paleomagnetic constraints from Hazara-Kashmir syntaxis in the western Himalaya. *Geological Society of America Bulletin*, *134*(5–6), 1109–1128. https://doi.org/10.1130/b36116.1

Jadoon, U. F., Zhao, Q., Huang, B., Yi, Z., Azeem, M. W., Lu, H., & Shah, S. A. (2024). A review of paleomagnetic constraints on the India-Asia collision: Paradoxes and perspectives. *Earth-Science Reviews*, *248*, 104628. https://doi.org/10.1016/j.earscirev.2023.104628

Jin, C., Liu, Q., Liang, W., Roberts, A. P., Sun, J., Hu, P., et al. (2018). Magnetostratigraphy of the Fenghuoshan Group in the Hoh Xil Basin and its tectonic implications for India–Eurasia collision and Tibetan plateau deformation. *Earth and Planetary Science Letters*, *486*, 41–53. https://doi.org/10.1016/j.epsl.2018.01.010

Jin, S., Sun, X., Jing, X., Zhang, Z., Zhang, X., & Yang, Z. (2023). A smaller greater India and a Middle-Early Eocene collision with Asia. *Geophysical Research Letters*, *50*(3), e2022GL101372. https://doi.org/10.1029/2022GL101372

Kapp, P., DeCelles, P. G., Gehrels, G. E., Heizler, M., & Ding, L. (2007). Geological records of the Lhasa-Qiangtang and Indo-Asian collisions in the Nima area of central Tibet. *Geological Society of America Bulletin*, *119*(7–8), 917–933. https://doi.org/10.1130/b26033.1

Kapp, P., Yin, A., Harrison, T. M., & Ding, L. (2005). Cretaceous-Tertiary shortening, basin development, and volcanism in central Tibet. *Geological Society of America Bulletin*, *117*(7), 865–878. https://doi.org/10.1130/b25595.1

Lallier, F., Antoine, C., Charreau, J., Caumon, G., & Ruiu, J. (2013). Management of ambiguities in magnetostratigraphic correlation. *Earth and Planetary Science Letters*, *371–372*, 26–36. https://doi.org/10.1016/j.epsl.2013.04.019

Li, S. (2024). Combined magnetostratigraphic and detrital zircon U-Pb dating of the tertiary Niubao Formation in the Bangong-Nujiang suture zone: Constraints on the amount of crustal shortening of the Tibetan Plateau since 60 Ma [Dataset]. *Zenodo*. https://doi.org/10.5281/zenodo.13170456

Li, S., Li, Y., Tan, X., Wang, C., Han, Z., Xiao, S., et al. (2022). New paleomagnetic results of the Upper Cretaceous to Lower Eocene sedimentary rocks from the Xigaze forearc basin and their tectonic implications. *Tectonophysics*, *837*, 229433. https://doi.org/10.1016/j.tecto.2022.229433

Li, S., van Hinsbergen, D. J. J., Najman, Y., Liu, J., Deng, C., & Zhu, R. (2020). Does pulsed Tibetan deformation correlate with Indian plate motion changes? *Earth and Planetary Science Letters*, *536*, 116144. https://doi.org/10.1016/j.epsl.2020.116144

19448007, 2024, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL111729 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Li, Y., Wang, C., Dai, J., Xu, G., Hou, Y., & Li, X. (2015). Propagation of the deformation and growth of the Tibetan–Himalayan orogen: A review. *Earth-Science Reviews*, *143*, 36–61. https://doi.org/10.1016/j.earscirev.2015.01.001

Liao, Z., & Bai, P. (2019). The discovery and significance of flora macrofossil in Niubao Formation of Bangongcuo-Nujiang river suture. *Yunnan Geology*, *38*(1), 10–14.

Liebke, U., Appel, E., Ding, L., Neumann, U., Antolin, B., & Xu, Q. (2010). Position of the Lhasa terrane prior to India-Asia collision derived from palaeomagnetic inclinations of 53 Ma old dykes of the Linzhou basin: Constraints on the age of collision and postcollisional shortening within the Tibetan plateau. *Geophysical Journal International*, *182*(3), 1199–1215. https://doi.org/10.1111/j.1365-246X.2010.04698.x

Lippert, P. C., van Hinsbergen, D. J. J., & Dupont-Nivet, G. (2014). Early cretaceous to present latitude of the central proto-Tibetan plateau: A paleomagnetic synthesis with implications for cenozoic tectonics, paleogeography, and climate of Asia. *Geological Society of America Special Paper*, *507*, 1–21. https://doi.org/10.1130/2014.2507(01

Lippert, P. C., Zhao, X., Coe, R. S., & Lo, C.-H. (2011). Palaeomagnetism and $^{40}Ar/^{39}Ar$ geochronology of upper Palaeogene volcanic rocks from central tibet: Implications for the central Asia inclination anomaly, the palaeolatitude of tibet and post-50 Ma shortening within Asia. *Geophysical Journal International*, *184*(1), 131–161. https://doi.org/10.1111/j.1365-246X.2010.04833.x

Liu, J., Su, T., Spicer, R. A., Tang, H., Deng, W., Wu, F., et al. (2019). Biotic interchange through lowlands of Tibetan Plateau suture zones during Paleogene. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *524*, 33–40. https://doi.org/10.1016/j.palaeo.2019.02.022

Liu, L., Liu, L., & Xu, Y.-G. (2021). Intermittent post-Paleocene continental collision in South Asia. *Geophysical Research Letters*, *48*(14), e2021GL094531. https://doi.org/10.1029/2021GL094531

Liu, X., Gao, R., Guo, X., & Ding, L. (2023). Detrital zircon U-Pb geochronology of the Lunpola basin strata constrains the Cenozoic tectonic evolution of central Tibet. *Gondwana Research*, *113*, 179–193. https://doi.org/10.1016/j.gr.2022.10.019

Ma, J., Yang, T., Peng, W., Bian, W., Wang, S., Han, F., et al. (2023). New Late Cretaceous paleomagnetic and geochronologic results from the southern Qiangtang Terrane: Contributions to the reliability of redbed paleomagnetic datasets from the Tibetan Plateau and shape of the Qiangtang Terrane. *Geological Society of America Bulletin*, *136*, 1769–1786. https://doi.org/10.1130/B37023.1

Martin, C. R., Jagoutz, O., Upadhyay, R., Royden, L. H., Eddy, M. P., Bailey, E., et al. (2020). Paleocene latitude of the Kohistan-Ladakh arc indicates multistage India-Eurasia collision. *Proceedings of the National Academy of Sciences of the United States of America*, *117*(47), 29487–29494. https://doi.org/10.1073/pnas.2009039117

McElhinny, M. W. (1964). Statistical significance of the fold test in paleomagnetism. *Geophysical Journal of the Royal Astronomical Society*, *8*(3), 338–340. https://doi.org/10.1111/j.1365-246x.1964.tb06300.x

McFadden, P. L. (1990). A new fold test for paleomagnetic studies. *Geophysical Journal International*, *103*(1), 163–169. https://doi.org/10.1111/j.1365-246X.1990.tb01761.x

McFadden, P. L., & McElhinny, M. W. (1990). Classification of the reversal test in paleomagnetism. *Geophysical Journal International*, *103*(3), 725–729. https://doi.org/10.1111/j.1365-246X.1990.tb05683.x

Meng, J., Gilder, S. A., Li, Y., Wang, C., & Liu, T. (2020). Expanse of greater India in the late cretaceous. *Earth and Planetary Science Letters*, *542*, 116330. https://doi.org/10.1016/j.epsl.2020.116330

Meng, J., Wang, C., Zhao, X., Coe, R., Li, Y., & Finn, D. (2012). India-asia collision was at 24 degrees N and 50 Ma: Palaeomagnetic proof from southernmost Asia. *Scientific Reports*, *2*(1), 925. https://doi.org/10.1038/srep00925

Murphy, M. A., Yin, A., Harrison, T. M., Dürr, S. B., Chen, Z., Ryerson, F. J., et al. (1997). Did the Indo-Asian collision alone create the Tibetan plateau? *Geology*, *25*(8), 719–722. https://doi.org/10.1130/0091-7613(1997)025<0719:dtiaca>2.3.co;2

Patzelt, A., Li, H., Wang, J., & Appel, E. (1996). Palaeomagnetism of Cretaceous to Tertiary sediments from southern Tibet: Evidence for the extent of the northern margin of India prior to the collision with Eurasia. *Tectonophysics*, *259*(4), 259–284. https://doi.org/10.1016/0040-1951(95)00181-6

Pusok, A. E., & Stegman, D. R. (2020). The convergence history of India-Eurasia records multiple subduction dynamics processes. *Science Advances*, *6*(19), eaaz8681. https://doi.org/10.1126/sciadv.aaz8681

Song, P., Ding, L., Zhao, T., Li, J., Yue, Y., & Xie, J. (2022). Paleomagnetism and geochronology of upper Eocene volcanic rocks from the western Qiangtang block: Constraints on the post-collisional shortening in western Tibet. *Global and Planetary Change*, *217*, 103953. https://doi.org/10.1016/j.gloplacha.2022.103953

Su, T., Spicer, R. A., Wu, F., Farnsworth, A., Huang, J., Del Rio, C., et al. (2020). A Middle Eocene lowland humid subtropical "Shangri-La" ecosystem in central Tibet. *Proceedings of the National Academy of Sciences of the United States of America*, *117*(52), 32989–32995. https://doi.org/10.1073/pnas.2012647117

Sun, J., Xu, Q., Liu, W., Zhang, Z., Xue, L., & Zhao, P. (2014). Palynological evidence for the latest Oligocene−early Miocene paleoelevation estimate in the Lunpola Basin, central Tibet. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *399*, 21–30. https://doi.org/10.1016/j.palaeo.2014.02.004

Sun, Z., Jiang, W., Li, H., Pei, J., & Zhu, Z. (2010). New paleomagnetic results of Paleocene volcanic rocks from the Lhasa block: Tectonic implications for the collision of India and Asia. *Tectonophysics*, *490*(3–4), 257–266. https://doi.org/10.1016/j.tecto.2010.05.011

Tan, X., Gilder, S., Kodama, K. P., Jiang, W., Han, Y., Zhang, H., et al. (2010). New paleomagnetic results from the Lhasa block: Revised estimation of latitudinal shortening across Tibet and implications for dating the India-Asia collision. *Earth and Planetary Science Letters*, *293*(3−4), 396–404. https://doi.org/10.1016/j.epsl.2010.03.013

Tauxe, L., & Kent, D. V. (2004). A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: Was the ancient magnetic field dipolar? *American Geophysical Union*, *145*, 101–115. https://doi.org/10.1029/145gm08

Tauxe, L., & Watson, G. S. (1994). The fold test-An eigen analysis approach. *Earth and Planetary Science Letters*, *122*(3–4), 331–341. https://doi.org/10.1016/0012-821X(94)90006-X

Tong, Y., Yang, Z., Gao, L., Wang, H., Zhang, X., An, C., et al. (2015). Paleomagnetism of upper cretaceous red-beds from the eastern Qiangtang block: Clockwise rotations and latitudinal translation during the India–Asia collision. *Journal of Asian Earth Sciences*, *114*, 732–749. https://doi.org/10.1016/j.jseaes.2015.08.016

Tong, Y., Yang, Z., Zheng, L., Yang, T., Shi, L., Sun, Z., & Pei, J. (2008). Early Paleocene paleomagnetic results from southern tibet, and tectonic implications. *International Geology Review*, *50*(6), 546–562. https://doi.org/10.2747/0020-6814.50.6.546

Torsvik, T. H., Van der Voo, R., Preeden, U., Niocaill, C. M., Steinberger, B., Doubrovine, P. V., et al. (2012). Phanerozoic polar wander, palaeogeography and dynamics. *Earth-Science Reviews*, *114*(3–4), 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007

Vandamme, D. (1994). A new method to determine paleosecular variation. *Physics of the Earth and Planetary Interiors*, *85*(1–2), 131–142. https://doi.org/10.1016/0031-9201(94)90012-4

Van der Voo, R., Spakman, W., & Bijwaard, H. (1999). Tethyan subducted slabs under India. *Earth and Planetary Science Letters*, *171*(1), 7–20. https://doi.org/10.1016/S0012-821X(99)00131-4

19448007, 2024, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL111729 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

van Hinsbergen, D. J. J., Kapp, P., Dupont-Nivet, G., Lippert, P. C., DeCelles, P. G., & Torsvik, T. H. (2011). Restoration of cenozoic deformation in Asia and the size of greater India. *Tectonics*, *30*(5), TC5003. https://doi.org/10.1029/2011tc002908

van Hinsbergen, D. J. J., Lippert, P. C., Dupont-Nivet, G., McQuarrie, N., Doubrovine, P. V., Spakman, W., & Torsvik, T. H. (2012). Greater India basin hypothesis and a two-stage cenozoic collision between India and Asia. *Proceedings of the National Academy of Sciences of the United States of America*, *109*(20), 7659–7664. https://doi.org/10.1073/pnas.1117262109

van Hinsbergen, D. J. J., Lippert, P. C., Li, S., Huang, W., Advokaat, E. L., & Spakman, W. (2019). Reconstructing Greater India: Paleogeographic, kinematic, and geodynamic perspectives. *Tectonophysics*, *760*, 69–94. https://doi.org/10.1016/j.tecto.2018.04.006

Wang, B., Zhou, J., Wen, T., & He, Z. (2009). Timing of terrestrial strata in Tibetan Nyima Basin and its significance. *Natural Gas Technology*, *3*(4), 21–24.

Wang, C., Dai, J., Zhao, X., Li, Y., Graham, S. A., He, D., et al. (2014). Outward-growth of the Tibetan plateau during the cenozoic: A review. *Tectonophysics*, *621*, 1–43. https://doi.org/10.1016/j.tecto.2014.01.036

Wang, C., Zhao, X., Liu, Z., Lippert, P. C., Graham, S. A., Coe, R. S., et al. (2008). Constraints on the early uplift history of the Tibetan Plateau. *Proceedings of the National Academy of Sciences of the United States of America*, *105*(13), 4987–4992. https://doi.org/10.1073/pnas.0703595105

Wang, S., Yang, T., Gao, F., Bian, W., Jin, J., Peng, W., et al. (2021). Paleomagnetic and geochronological results of the Risong Formation in the western Lhasa terrane: Insights into the Lhasa-Qiangtang collision and stratal age. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *586*, 110778. https://doi.org/10.1016/j.palaeo.2021.110778

Wei, W., Lu, Y., Xing, F., Liu, Z., Pan, L., & Algeo, T. J. (2017). Sedimentary facies associations and sequence stratigraphy of source and reservoir rocks of the lacustrine Eocene Niubao Formation (Lunpola Basin, central Tibet). *Marine and Petroleum Geology*, *86*, 1273–1290. https://doi.org/10.1016/j.marpetgeo.2017.07.032

Westerweel, J., Roperch, P., Licht, A., Dupont-Nivet, G., Win, Z., Poblete, F., et al. (2019). Burma Terrane part of the Trans-Tethyan arc during collision with India according to palaeomagnetic data. *Nature Geoscience*, *12*(10), 863–868. https://doi.org/10.1038/s41561-019-0443-2

Xia, D., & Liu, S. (2008). *Stratigraphy (lithostratic) of Xizang autonomous region*. China University of Geosciences Press.

Xiao, S., Li, Y., He, J., Wang, C., He, H., Cheng, J., et al. (2022). Late Permian rift-related volcanic rocks from the Zhongba terrane, southern tibet: Implications for the opening of the Neotethys. *International Geology Review*, *65*(10), 1–14. https://doi.org/10.1080/00206814.2022.2103847

Xiong, Z., Liu, X., Ding, L., Farnsworth, A., Spicer, R. A., Xu, Q., et al. (2022). The rise and demise of the Paleogene central Tibetan valley. *Science Advances*, *8*(6), eabj0944. https://doi.org/10.1126/sciadv.abj0944

Xu, Y., Tan, X., Li, S., Li, Y., Ran, B., Han, Y., et al. (2022). Paleomagnetism of the Greater Indian passive margin sediments from the Upper Cretaceous succession: Evidence for long-delayed remagnetizations and implication for the India-Asia collision. *Journal of Asian Earth Sciences*, *229*, 105165. https://doi.org/10.1016/j.jseaes.2022.105165

Yang, G., Wang, Z., Chen, J. W., Yan, D., & Sun, B. (2016). *Equisetum cf. oppositum* (Equisetaceae) from the Paleocene–Eocene of Tibet in southwestern China and its paleoenvironmental implications. *Arabian Journal of Geosciences*, *9*(20), 1–10. https://doi.org/10.1007/s12517-016-2777-z

Yang, T., Jin, J., Bian, W., Ma, Y., Gao, F., Peng, W., et al. (2019). Precollisional latitude of the northern Tethyan Himalaya from the Paleocene redbeds and its implication for greater India and the India-Asia collision. *Journal of Geophysical Research: Solid Earth*, *124*(11), 10777–10798. https://doi.org/10.1029/2019jb017927

Yi, Z., Huang, B., Chen, J., Chen, L., & Wang, H. (2011). Paleomagnetism of early Paleogene marine sediments in southern Tibet, China: Implications to onset of the India-Asia collision and size of Greater India. *Earth and Planetary Science Letters*, *309*, 153–165. https://doi.org/10.1016/j.epsl.2011.07.001

Yi, Z., Liang, Y., Zhao, J., Yan, Y., Chen, L., & Tang, X. (2016). Paleogeography of the northern margin of Indian continent prior to its collision: An investigation of the late cretaceous limestones in south tibet. *Acta Geologica Sinica*, *90*(11), 3282–3292. (in Chinese with English abstract).

Yi, Z., Wang, T., Meert, J. G., Zhao, Q., & Liu, Y. (2021). An initial collision of India and Asia in the equatorial humid belt. *Geophysical Research Letters*, *48*(9), e2021GL093408. https://doi.org/10.1029/2021GL093408

Yin, A., & Harrison, T. M. (2000). Geologic evolution of the Himalayan-Tibetan orogen. *Annual Review of Earth and Planetary Sciences*, *28*(1), 211–280. https://doi.org/10.1146/annurev.earth.28.1.211

Yin, S., Zhang, Z., Huang, Y., Zhu, Y., Zhang, S., Han, Z., et al. (2021). The chemo-stratigraphic analysis for the Paleocene–Eocene Niubao Formation on the section in southern Xiede village, the Nyima basin, Xizang (tibet). *Geological Review*, *67*(3), 668–681.

Yuan, J., Deng, C., Yang, Z., Krijgsman, W., Thubtantsering, Qin, H., et al. (2022). Triple-stage India-Asia collision involving arc-continent collision and subsequent two-stage continent-continent collision. *Global and Planetary Change*, *212*, 103821. https://doi.org/10.1016/j.gloplacha.2022.103821

Yuan, J., Deng, C., Yang, Z., Krijgsman, W., Thubtantsering, Qin, H., et al. (2023). New paleomagnetic data from the central Tethyan Himalaya refine the size of Greater India during the Campanian. *Earth and Planetary Science Letters*, *622*, 118422. https://doi.org/10.1016/j.epsl.2023.118422

Yuan, J., Yang, Z., Deng, C., Krijgsman, W., Hu, X., Li, S., et al. (2020). Rapid drift of the Tethyan Himalaya terrane before two-stage India-Asia collision. *National Science Review*, *0*, 1–13. https://doi.org/10.1093/nsr/nwaa173

Zeng, S., Chen, W., Fu, X., Liu, J., Li, J., & Chou, J. (2020). The discovery of Paleogene insect fossils in eastern Nima Basin, Tibet, and its implication for paleoclimate. *Geological Bulletin of China*, *39*(7), 1121–1124.

Zhang, B., Li, Y., & Hu, X. (2017). Paleomagnetic results from Jurassic–Cretaceous strata in the Chuangde area of southern Tibet constrain the nature and timing of the India-Asia collision system. *Chinese Science Bulletin*, *62*(4), 298–311. (in Chinese with English abstract). https://doi.org/10.1360/N972016-00767

Zhang, J., Xiao, W., Wakabayashi, J., Windley, B. F., & Han, C. (2022). A fragment of Argoland from east Gondwana in the NE Himalaya. *Journal of Geophysical Research: Solid Earth*, *127*(2), e2021JB022631. https://doi.org/10.1029/2021JB022631

Zhang, W., Fang, X., Zhang, T., Song, C., & Yan, M. (2020). Eocene rotation of the northeastern central Tibetan plateau indicating stepwise compressions and eastward extrusions. *Geophysical Research Letters*, *47*(17), e2020GL088989. https://doi.org/10.1029/2020gl088989

Zhang, X., Gelin, U., Spicer, R. A., Wu, F., Farnsworth, A., Chen, P., et al. (2022). Rapid Eocene diversification of spiny plants in subtropical woodlands of central Tibet. *Nature Communications*, *13*(1), 1–10. https://doi.org/10.1038/s41467-022-31512-z

Zhao, Z., Wu, Z., Yang, Y., & Ji, C. (2020). Establishing the chronostratigraphic framework of the continental red beds in Central Qiangtang Basin: Constrained by zircon U-Pb ages. *Geological Review*, *66*(5), 1155–1170.

19448007, 2024, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL111729 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References From the Supporting Information

Ahmad, M. N., Yoshida, M., & Fujiwara, Y. (2000). Paleomagnetic study of Utror volcanic formation: Remagnetizations and postfolding rotations in Utror area, Kohistan arc, northern Pakistan. *Earth Planets and Space*, *52*(6), 425–436. https://doi.org/10.1186/BF03352254

Ahmad, M. N., Fujiwara, Y., & Paudel, L. P. (2001). Remagnetization of igneous rocks in Gupis area of Kohistan arc, northern Pakistan. *Earth Planets and Space*, *53*(5), 373–384. https://doi.org/10.1186/BF03352394

Bian, W., Yang, T., Jiang, Z., Jin, J., Gao, F., Wang, S., et al. (2020). Paleomagnetism of the Late Cretaceous red beds from the far western Lhasa terrane: Inclination discrepancy and tectonic implications. *Tectonics*, *39*(8), e2020TC006280. https://doi.org/10.1029/2020TC006280

Cao, Y., Sun, Z., Liu, D., Zhang, L., Ye, X., Zheng, Y., & He, X. (2017a). Late Cretaceous paleomagnetic results in the Nyima area from the northern margin of the Lhasa block and its tectonic implications. *Acta Petrologica Sinica*, *33*, 3989–3998. (in Chinese with English abstract).

Cao, Y., Sun, Z., Li, H., Pei, J., Jiang, W., Xu, W., et al. (2017b). New Late Cretaceous paleomagnetic data from volcanic rocks and red beds from the Lhasa terrane and its implications for the paleolatitude of the southern margin of Asia prior to the collision with India. *Gondwana Research*, *41*, 337–351. https://doi.org/10.1016/j.gr.2015.11.006

Cogné, J. P. (2003). PaleoMac: A Macintosh™ application for treating paleomagnetic data and making plate reconstructions [Software]. *Geochemistry, Geophysics, Geosystems*, *4*(1), 1007. https://doi.org/10.1029/2001GC000227

Gilder, S., Chen, Y., & Sen, S. (2001). Oligo-Miocene magnetostratigraphy and rock magnetism of the Xishuigou section, Subei (Gansu Province, western China) and implications for shallow inclinations in central Asia. *Journal of Geophysical Research*, *106*(B12), 30505–30521. https://doi.org/10.1029/2001jb000325

Gradstein, F., Ogg, J., & Smith, A. (2004). *A geologic time scale 2004*. Cambridge University Press.

Jadoon, U. F., Huang, B., Zhao, Q., Shah, S. A., & Rahim, Y. (2021b). Remagnetization of Jutal dykes in Gilgit area of the Kohistan Island arc: Perspectives from the India-Asia collision. *Geophysical Journal International*, *226*(1), 33–46. https://doi.org/10.1093/gji/ggab091

Jiang, Z., Liu, Q., Dekkers, M. J., Tauxe, L., Qin, H., Barrón, V., & Torrent, J. (2015). Acquisition of chemical remanent magnetization during experimental ferrihydrite-hematite conversion in Earth-like magnetic field–implications for paleomagnetic studies of red beds. *Earth and Planetary Science Letters*, *428*, 1–10. https://doi.org/10.1016/j.epsl.2015.07.024

Kirschvink, J. L. (1980). The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal International*, *62*(3), 699–718. https://doi.org/10.1111/j.1365-46X.1980.tb02601.x

Koymans, M. R., Langereis, C. G., Pastor-Galán, D., & van Hinsbergen, D. J. J. (2016). Paleomagnetism.org: An online multi-platform open source environment for paleomagnetic data analysis [Software]. *Computers and Geosciences*, *93*, 127–137. https://doi.org/10.1016/j.cageo.2016.05.007

Liang, Y., Huang, B., Yi, Z., Zhang, Y., Yang, Y., & Zhang, D. (2017). Reconnaissance of pre-collisional paleolatitudes of the southern margin of Eurasia: New paleomagnetic results from Upper Cretaceous red beds in the Linzhou basin, Tibet. *Chinese Journal of Geophysics*, *60*(5), 1811–1824. (in Chinese with English abstract). https://doi.org/10.6038/cjg20170517

Liu, Y., Zhou, Y., Cheng, X., Ye, Y., Wei, B., Jiang, N., et al. (2022). Paleomagnetism of the late cretaceous Jingzhushan Formation red beds on the northern margin of the Lhasa terrane in the Qinghai-Tibet Plateau and its tectonic significance. *Progress in Geophysics*, *37*(6), 2275–2290. (in Chinese with English abstract).

Ma, Y., Yang, T., Bian, W., Jin, J., Wang, Q., Zhang, S., et al. (2017). Paleomagnetic and geochronologic results of latest Cretaceous lava flows from the Lhasa terrane and their tectonic implications. *Journal of Geophysical Research: Solid Earth*, *122*(11), 8786–8809. https://doi.org/10.1002/2017JB014743

Ma, Y., Yang, T., Bian, W., Jin, J., Wang, Q., Zhang, S., et al. (2018). A stable southern margin of Asia during the Cretaceous: Paleomagnetic constraints on the Lhasa-Qiangtang collision and the maximum width of the Neo-Tethys. *Tectonics*, *37*(10), 3853–3876. https://doi.org/10.1029/2018TC005143

Ma, Y., Wang, Q., Yang, T., Ou, Q., Zhang, X., Dan, W., et al. (2022). Location of the Lhasa terrane in the Late Cretaceous and its implications for crustal deformation. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *588*, 110821. https://doi.org/10.1016/j.palaeo.2021.110821

Pozzi, J. P., Westphal, M., Zhou, Y., Xing, L., & Chen, X. (1982). Position of the Lhasa block, south tibet, during the late cretaceous. *Nature*, *297*(5864), 319–321. https://doi.org/10.1038/297319a0

Qu, Y., Wang, Y., Zhang, S., Wang, Z., Lv, P., & Duan, J. (2004). New results and major progress in regional geological survey of the Toiba District Sheet. *Geological Bulletin of China*, *23*(5), 492–497.

Qu, Y. (2011). Regional geological survey report of the People's Republic of China–Toiba District (H45C001004) sheet in Xizang (scale 1: 250000). In *strata*. Geological Publishing House.

Sun, Z., Pei, J., Li, H., Xu, W., Jiang, W., Zhu, Z., et al. (2012). Palaeomagnetism of late Cretaceous sediments from southern Tibet: Evidence for the consistent palaeolatitudes of the southern margin of Eurasia prior to the collision with India. *Gondwana Research*, *21*(1), 53–63. https://doi.org/10.1016/j.gr.2011.08.003

Tong, Y., Yang, Z., Pei, J., Wang, H., Xu, Y., & Pu, Z. (2017). Paleomagnetism of the upper cretaceous red-beds from the eastern edge of the Lhasa terrane: New constraints on the onset of the India-Eurasia collision and latitudinal crustal shortening in southern Eurasia. *Gondwana Research*, *48*, 86–100. https://doi.org/10.1016/j.gr.2017.04.018

Tong, Y., Yang, Z., Pei, J., Li, J., Jin, S., Hou, L., et al. (2022). New Late Cretaceous paleomagnetic results from the Lhasa terrane and their implications for the suturing of India and Eurasia and the closure of the Neo-Tethys Ocean. *Geological Society of America Bulletin*, *134*(11–12), 3242–3257. https://doi.org/10.1130/b36310.1

Westphal, M., Pozzi, J. P., Zhou, Y., Xing, L., & Chen, X. (1983). Palaeomagnetic data about southern tibet (Xizang)—I. The cretaceous formations of the Lhasa block. *Geophysical Journal International*, *73*(2), 507–521. https://doi.org/10.1111/j.1365-246X.1983.tb03327.x

Yang, T., Ma, Y., Zhang, S., Bian, W., Yang, Z., Wu, H., et al. (2015). New insights into the India-Asia collision process from Cretaceous paleomagnetic and geochronologic results in the Lhasa terrane. *Gondwana Research*, *28*(2), 625–641. https://doi.org/10.1016/j.gr.2014.06.010

Zijderveld, J. D. A. (1967). A. C. demagnetization of rocks: Analysis of results. In D. W. Collinson, K. M. Creer, & S. K. Runcorn (Eds.), *Methods in paleomagnetism* (pp. 254–286). Elsevier.

19448007, 2024, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL111729 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License